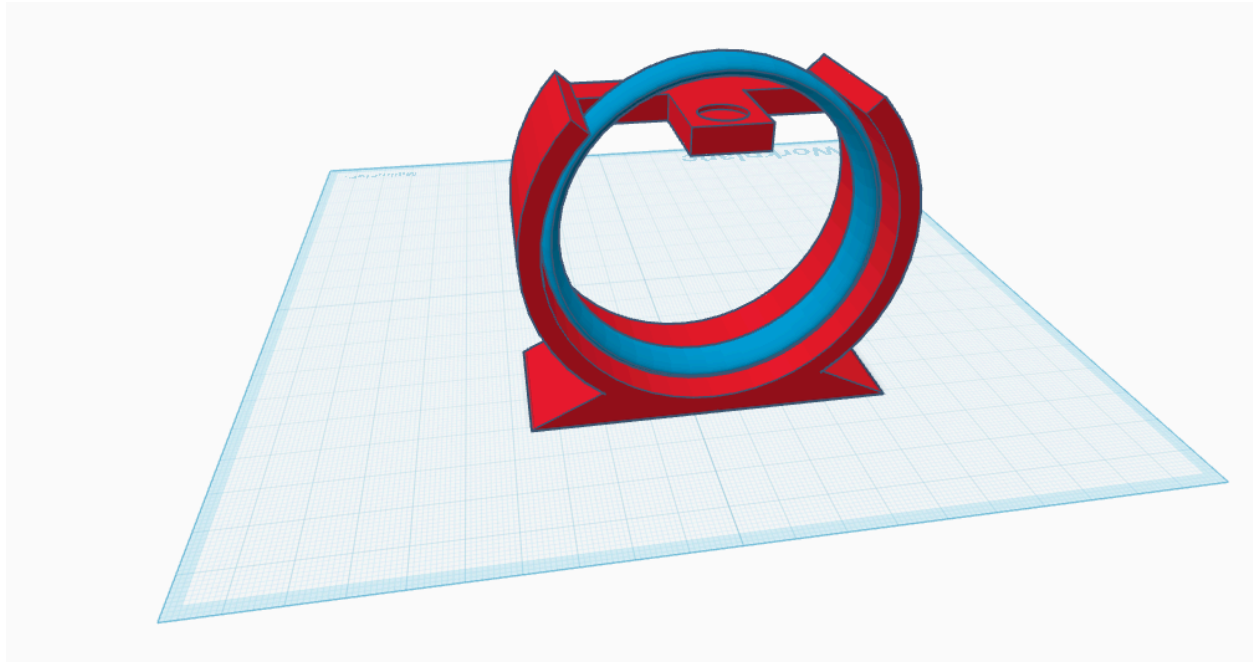


One-Handed

Portal



How it works

The design allows the daycare members to attach the wristband to the child's wrist using only one hand. This allows the members to potentially multitask and perform other morning duties as necessary. The design allows the wristband to be placed into a circular "C" shaped structure, with the "male" part of the snap sitting in a divot in the tab. This allows the wristband to stay in position, making the application process easier. The tab also acts as a tool for ensuring the wristband is not too tight. Then, with the wrist still inside the structure, the staff member can cut off any excess as required. The wrist can then be removed and the wristband will be attached.

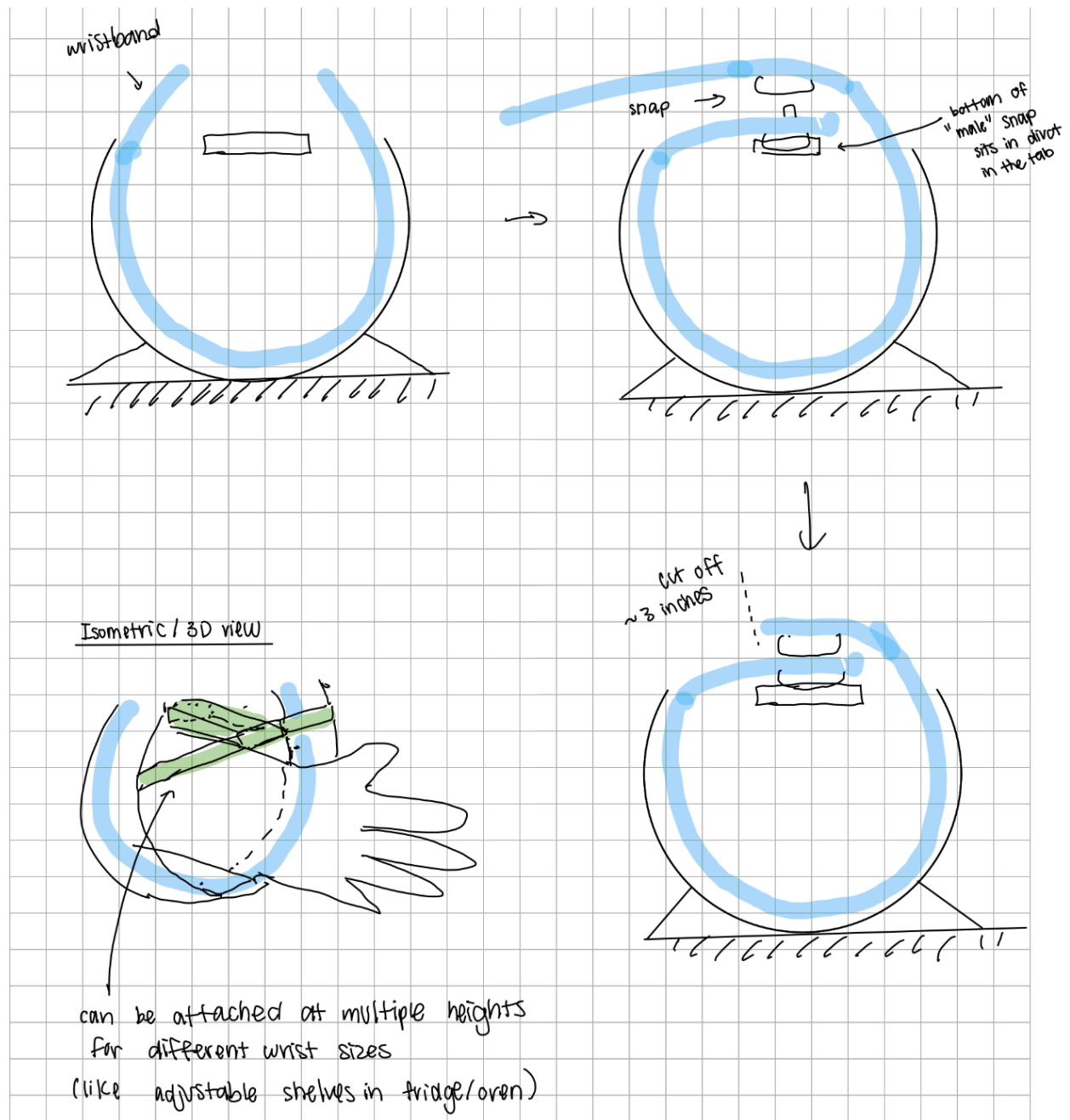
Requirements + Evaluation Criteria

The design shall require ≤ 60 seconds of active staff time to set up before distribution/wristband application begins.	✓	- Minimal set up time required, wristband is easily placed into slot
The wristband application process shall require less than 5 steps .	✓	- Application process takes 5 steps

The design shall allow a wristband to be applied in $\leq$ 34 seconds .	✓	- Quick wristband application
The design shall require $\leq$ 5-10 minutes to prepare 90 wristbands before set-up.	✓	- Minimal preparation required
The design shall keep the verification time of each kid to be in $\leq$ 5 seconds .	✓	- Yes
The design shall stay reliable after being used for 5 years .	✓	- Yes
The design shall ensure that each application can be completed using a single simple action (e.g. place, fasten, remove access)	✓	- Yes
The design shall allow the wristband application process to be understood and executed by users aged 4+	✓	- Simple steps should be easy to understand
The design shall accommodate users within the 5th–95th percentile anthropometric hand size range.	✓	- Yes, still working on adjustability
The wristband shall allow a clearance of 10–20 mm between the wristband and the child's wrist, equivalent to the two-finger spacing guideline, to prevent excessive compression while maintaining retention.	✓	- Yes, built into the design
The wristband shall remain securely attached during normal daycare activities for up to 1 week.	✓	- Yes
Any wristband and application	✓	- Yes

mechanism shall have all edges fulfill the UL1439 Sharp Edge Test standard.		
Any wristband application mechanism shall have 0 accessible pinch points and entrapment hazards during any step in the usage process.	✓	- Yes, no pinch points
Coverings of mechanical points and other hazards must withstand 66.8N of tension and 133.4N of compression force.	✓	- Yes
Any wristband application process shall apply a force of no more than 15N to users wrists.	✓	- Yes, the tab distributes the applied force more equally
Any manual steps in the application process shall require no more than 22N of force to complete.	✓	- Yes, same force as original wristband

Sketch:



Research

Medca Bracelet Buddy

Recommendation: C-shaped tool to allow one-handed wristband application



Source: <https://www.amazon.ca/Bracelet-Jewelry-Fastening-Unfasten-Bracelets/dp/B08DJB38DC>

Properties of tool that contribute to requirements and evaluation criteria

- 5 step process
- speed - quick to use, minimal set-up required
- adjustable for all wrist sizes
- minimal force is applied to the user's wrist

Limitations

- has a potential pinching point (the clasp), making it unsafe for children
- potentially difficult to use and hard to use for people with limited fine motor skills

Primary Advantage

- Simple to use, minimal steps required
- Provides “two fingers” measurement to ensure wristband is not too tight

Primary Disadvantages

- Difficult to adjust for all wrist sizes
- Stakeholder meeting - they do not see the value in a one-handed approach
- Force applied to snap is still applied to wrist, albeit not directly

Biggest Questions

- How to make the design adjustable for all wrist sizes? Moving the tab up/down?

Next Steps

- Finding a way for the design to be able to accommodate all wrist sizes
- Making the tab part stronger and easier for it to hold onto the snap

Automatic

Magnetomatic

How it works

This design allows the wristband application to be finished with minimum or no extra work required for the staff, since the staff would not need to help each kid to wear the wristbands. It can also reduce the effects of kid's imparity during the process. The overall efficiency will therefore be improved and time for the morning preparation period can be reduced.

The design mainly contains a bracelet with two magnetic sides and a metal skeleton. The two magnets are attached to the skeleton before being applied. The user needs to put one of the hands between the bracelet and the skeleton first, and move his or her hand toward the bracelet. By exerting force to the bracelet, the magnet on the top will detach from the skeleton and fall down due to gravity. It can automatically attach to the magnet at the bottom and therefore wrap the wrist. Once the user keeps moving the hand apart, the second magnet will detach from the skeleton.

If we replace the classical magnets with electromagnets, kids will not be able to unfasten the bracelet until the staff unlock it, making the design also reusable.

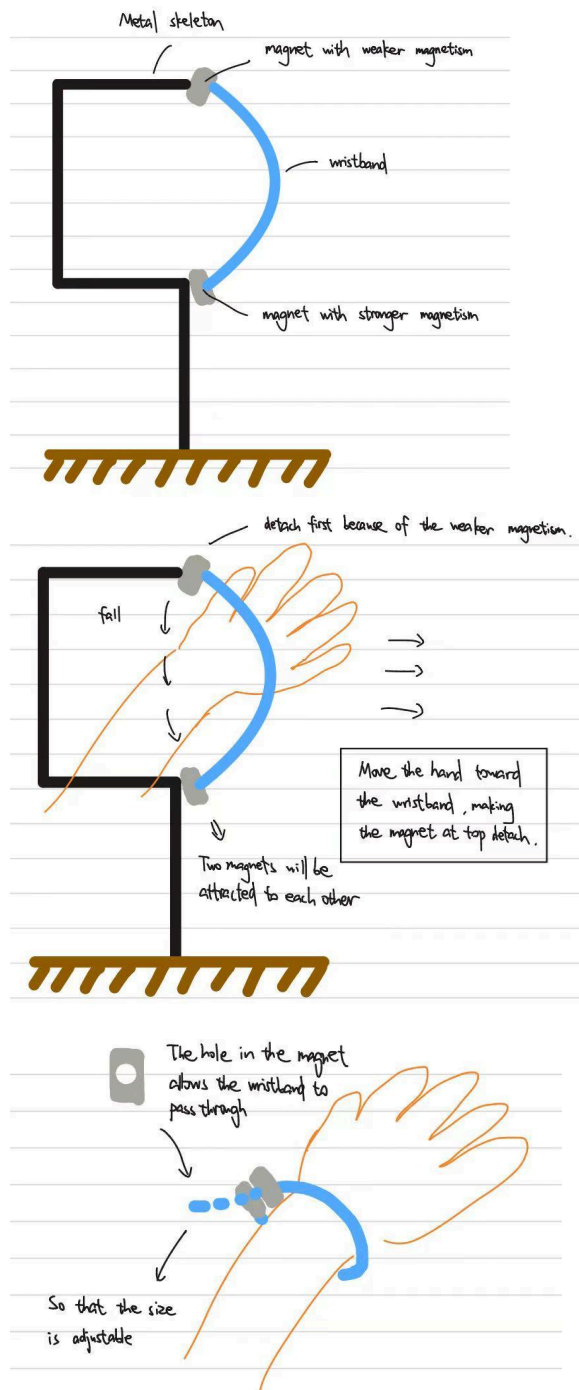
Requirements + Evaluation Criteria

The design shall require ≤ 60 seconds of active staff time to set up before distribution/wristband application begins.	✓	- The skeleton can be easily set up
The wristband application process shall require less than 5 steps .	✓	- It only contains 3 steps: setting up the skeleton, attaching the wristband, and moving the hand.
The design shall allow a wristband to be applied in ≤ 34 seconds .	✓	- Attaching + detaching process will only use a short period of time.
The design shall require $\leq 5-10$ minutes to prepare 90 wristbands before set-up.	✓	- Minimal preparation required

The design shall keep the verification time of each kid to be in ≤ 5 seconds.	✓	- Yes
The design shall stay reliable after being used for 5 years.	✓	- Yes if using a proper material
The design shall ensure that each application can be completed using a single simple action (e.g. place, fasten, remove access)	✓	- Yes
The design shall allow the wristband application process to be understood and executed by users aged 4+	✓	- The actions are easy to follow, but might be affected by some potential unpredictable motions.
The design shall accommodate users within the 5th–95th percentile anthropometric hand size range.	✓	- The sizes are adjustable.
The wristband shall allow a clearance of 10–20 mm between the wristband and the child's wrist, equivalent to the two-finger spacing guideline, to prevent excessive compression while maintaining retention.	✓	- The sizes are adjustable
The wristband shall remain securely attached during normal daycare activities for up to 1 week.	✓	- Yes if using electromagnets.
Any wristband and application mechanism shall have all edges fulfill the UL1439 Sharp Edge Test standard.	✓	- The shape of the magnets can be modified
Any wristband application mechanism shall have 0 accessible pinch points and entrapment hazards during any	✓	- There might be potential risks on this problem, but can be avoided by modifying the shape of the magnets or adding some soft foams or other materials between the magnets.

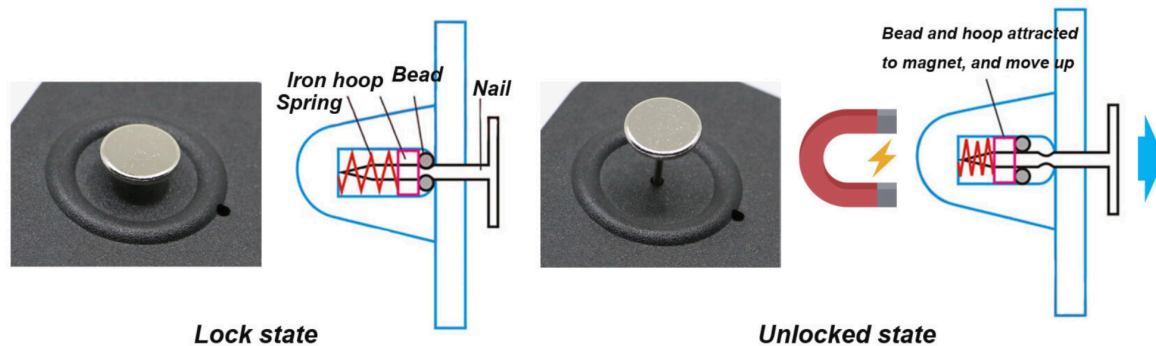
step in the usage process.		
Coverings of mechanical points and other hazards must withstand 66.8N of tension and 133.4N of compression force.	✓	- Yes
Any wristband application process shall apply a force of no more than 15N to users wrists.	✓	- Yes
Any manual steps in the application process shall require no more than 22N of force to complete.	✓	- Yes

Sketch:



Research

Electromagnetic Security Tag



Recommendation: Application of electromagnets when locking and unlocking the wristband.

Source: <https://www.magnet-sdm.com/brief-introduction-security-tag-remover/>

Properties of [object] that contribute to requirements and evaluation criteria

- Requires less than 5 steps to complete, the whole process is less than 34 seconds
- Adjustable to all wrist sizes
- Reusable and durable
- Force applied to the user's wrist is adjustable so can be minimized

Limitations

- Magnets are heavy, making the wristband heavy
- Difficulties for young kids to wear wristbands by themselves
- The process when the magnet fall down might hit the wrist or hand of the kid, might cause potential safety issues

Primary Advantage

- Automatically applied, which could minimize the work load for the staff
- If using electromagnetic design, the wristband would then be reusable

Primary Disadvantages

- Kids' behaviours are unpredictable. They might be unwilling to cooperate, making it hard for some of the young students to wear the wristband properly.

Biggest Questions

- How to make sure that the magnets are attached properly every time?

Next Steps

- Brainstorming some methods to make this design better or more reliable;
- Thinking about the suitable materials for the wristband;
- Considering a method to reduce the overall weight of the wristband due to the heavy magnets.

Simplify

Knobby

How it works

This design takes the approach of creating a new wristband that better fulfills the stakeholder's needs. The band is made from a sturdy but comfortable material that is cognizant of children's sensory needs as a primary stakeholder. The ends of the wristband meet at a round mechanism, not larger than the size of a watch. This is also made of a safe material that is comfortable at where it's in contact with the skin. When rotated clockwise, this mechanism tightens the loop part of the bracelet. If squeezed and rotates counter clockwise, it "pops out" and allows the user to loosen the loop. This acts as a sturdy and durable child lock that also has space for an identification mechanism on top.

Requirements

The design shall require ≤ 60 seconds of active staff time to set up before distribution/wristband application begins.	✓	Yes - Same amount of time as current wristbands
The wristband application process shall require less than 5 steps .	✓	Yes - 2 steps, 5 is readjusted
The design shall allow a wristband to be applied in ≤ 34 seconds .	✓	Yes
The design shall require $\leq 5-10$ minutes to prepare 90 wristbands before set-up.	✓	Yes - just organize, no unique preparations needed
The design shall keep the verification time of each kid to be in ≤ 5 seconds .	✓	Yes - large and clear
The design shall stay reliable after being used for 5 years .	✓	Yes - based off of strength of reference designs

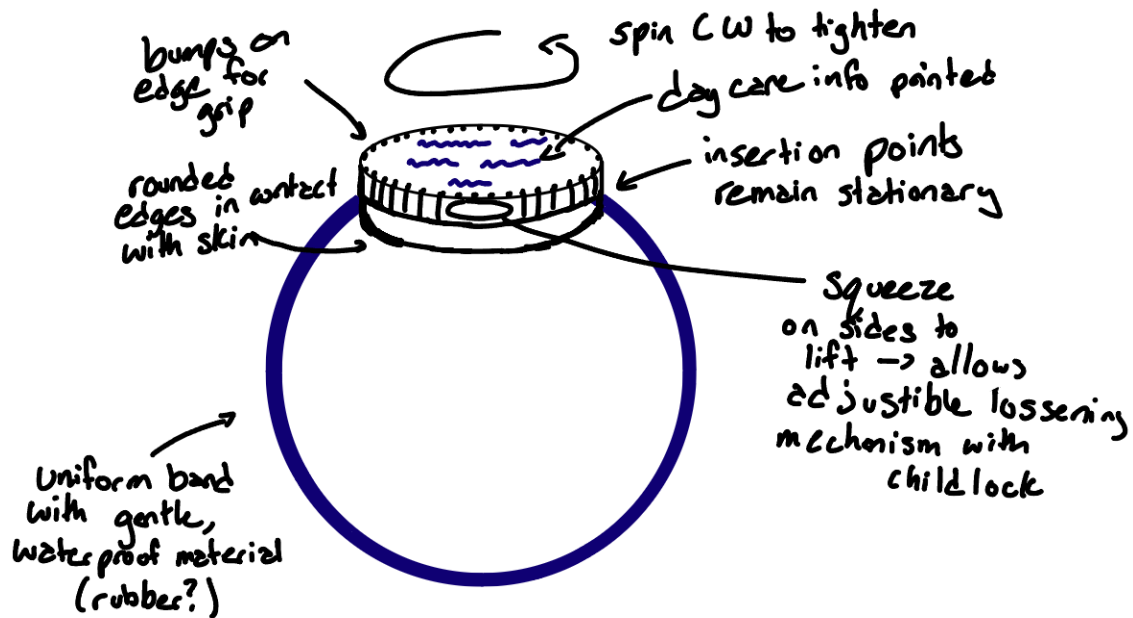
The design shall ensure that each application can be completed using a single simple action (e.g. place, fasten, remove access)	✓	Yes - rotations, squeeze if its needed to bypass childlock
The design shall allow the wristband application process to be understood and executed by users aged 4+	✓	Yes - children can spin a knob
The design shall accommodate users within the 5th–95th percentile anthropometric hand size range.	✓	Yes - fully customizable sizing
The wristband shall allow a clearance of 10–20 mm between the wristband and the child’s wrist, equivalent to the two-finger spacing guideline, to prevent excessive compression while maintaining retention.	✓	Yes - fully customizable sizing
The wristband shall remain securely attached during normal daycare activities for up to 1 week.	✓	Yes - durable and not removable by kids
Any wristband and application mechanism shall have all edges fulfill the UL1439 Sharp Edge Test standard.	✓	Yes - rounded edges
Any wristband application mechanism shall have 0 accessible pinch points and entrapment hazards during any step in the usage process.	✓	- Yes - all internal
Coverings of mechanical points and other hazards must withstand 66.8N of tension and 133.4N of compression force.	✓	- Yes - mechanical strength similar to that of a watch
Any wristband application	✓	- Yes - depends on customizable tightness

process shall apply a force of no more than 13.5N to users wrists.

Any manual steps in the application process shall require no more than 22N of force to complete.

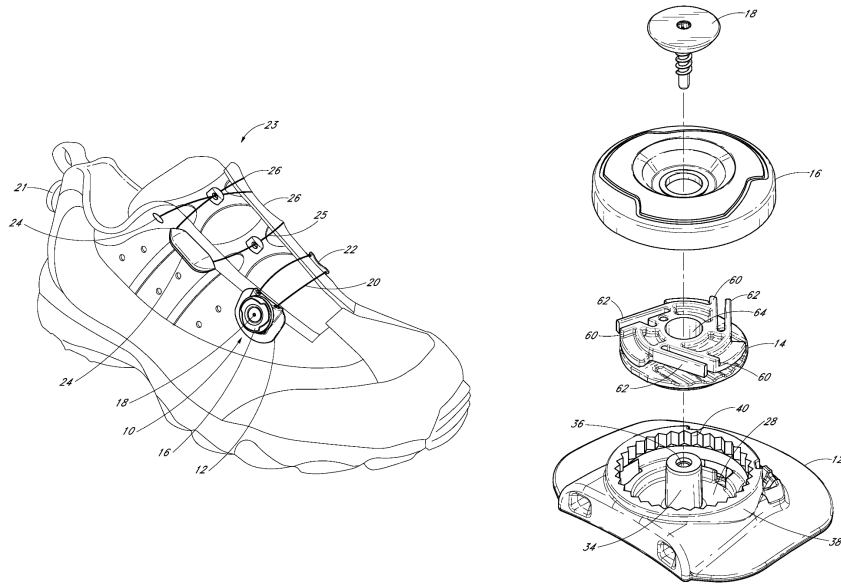


- Yes - dial is relatively easy to spin based off of references



Research

BOA® Fit System - <https://www.boafit.com/en-us>



Recommendation: Reel based lacing system

Source: <https://patents.google.com/patent/US8468657B2/en>

Patent: US8468657B2

Location: United States

Inventor: Mark S. Soderberg, Michael Joseph Nickel, Sean Cavanagh, James Paul Goodman

Assignee: Boa Technology Inc

Status: Active - set to expire on 2031-06-18

Properties of system that contribute to requirements and evaluation criteria

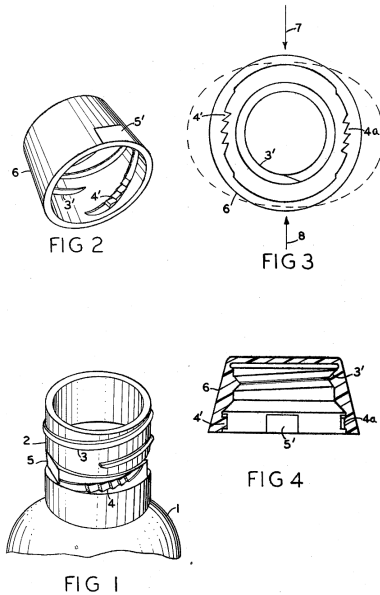
- Speed - minimum of two steps to slide on and spin to tightened
 - Reusable, durable
 - Freely able to tighten as required - easily usable for different sizes and adjustable to sensory needs
 - Moving components completely covered, child lock
-

Limitations

- Intended for steel laces on shoes, different materials/application may not work as effectively
- Locking mechanism is easy to bypass

Medicine Cap Lid Child Lock

U.S. Patent Jun. 2, 1981 Sheet 1 of 2 4,270,664



Recommendation: One piece child resistant cap

Source: <https://patents.google.com/patent/US4270664>

Patent: US4270664A

Location: United States

Inventor: Caetano Buono

Assignee: Van Blarcom Closures Inc

Status: Expired in 1999

Properties of system that contribute to requirements and evaluation criteria

- Speed - minimum of 2 steps
- Reusable - allows for adjustments to be not disposable
- Simple to use for adults
- Safe and effective child lock

Limitations

- Intended for medicine bottle caps, may not be entirely implementable in this context.
- Uncomfortable for sensory needs of kids.

Primary Advantage

- Only quick, simple actions needed to apply
- Easily adjustable to different children
- Can be removed and reused
-

Primary Disadvantages

- Still may require staff assistance and energy
- Potential for temporary pinching if applied hastily
- Relatively large (size of childrens analog watch) excess material - may be uncomfortable
-

Biggest Questions

- How small can we make it in order to fit all complex mechanics?
- Materials selection: determine what is strong enough but doesn't risk agitating the skin
- What the cost may be - does it genuinely outweigh the null?

Next Steps

- If selected, must reframe and add requirements pertaining to creating a wristband, not an application mechanism
- Look into failure cases to ensure safety
- Develop high fidelity prototypes or CAD designs to test various requirements and ECs